

Scanner Identification Using Sensor Pattern Noise

Group 28

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Introduction & Objective

- Problem: Digitally scanned images can originate from multiple scanner devices. Verifying which scanner produced a given image is critical for forensic authentication and document integrity.
- Objective: Identify the source scanner of a scanned image using the scanner's inherent noise pattern as a unique device fingerprint.
- Core Approach: Extract noise residuals using wavelet denoising, build per-scanner reference patterns, and classify using 2D correlation, 1D row correlation, and an SVM classifier.
- Dataset: 4 real flatbed scanners (Epson 4490, HP ScanJet 6300c ×2, HP ScanJet 8250), 160 images at 1200 DPI, and 24,270 total patches.
- Outcome: SVM accuracy reached 88.44%, while the 2D baseline achieved 67.19% and the 1D row-correlation baseline achieved 70.31%.

Proposed Identification Pipeline

- Dataset Loading: Traverse scanner_dataset/scanner_name/dpi/ folders to collect TIFF paths with scanner labels (s1–s4) and keep only the 1200 DPI images used in the main experiment.
- Patch Extraction: Extract non-overlapping 1024×768 patches from each valid TIFF scan to obtain 24,270 patches in total.
- Wavelet Denoising: Apply 4-level db8 wavelet denoising with soft thresholding and MAD-based noise estimation.
- Residual Extraction: Subtract the denoised patch from the original patch to isolate the sensor noise residual.
- Feature Extraction: Build a 16-dimensional feature vector from row-pattern and correlation statistics.
- Classification: Use group-aware train/test splitting, GroupKFold cross-validation, and an RBF-kernel SVM, with 2D and 1D correlation baselines for comparison.

Dataset & Sample Images

- Scanners Used: s1_epson_4490, s2_hp_scanjet_6300c_1_SG9CO270W5, s3_hp_scanjet_6300c_2, s4_hp_scanjet_8250.
- Format: High-resolution TIFF images scanned at native DPI, stored under per-scanner folders.
- Total Images Used: 160 images at 1200 DPI for the main experiment, selected from a larger collection of 1,583 TIFF files.
- Total Patches: 24,270 non-overlapping sub-images of size 1024×768 pixels.
- Class Note: The two HP 6300c units are the hardest pair because they are physically very similar.
- Sample Image: The sample TIFF shown here is the raw grayscale scan used to verify loading and display.

Residual Extraction — Patch / Denoised / Noise

- Patch: Non-overlapping 1024×768 grayscale region from the original TIFF scan.
- Denoised: Output of 4-level db8 wavelet denoising, which suppresses most scene content while preserving structured residual behavior.
- Noise Residual: Appears as a low-amplitude but structured residual; this is the scanner fingerprint used for later classification.
- Mathematical Step: $\text{Noise}(x, y) = \text{Patch}(x, y) - \text{Denoised}(x, y)$
- Processing Stats: All 24,270 patches were processed successfully with 0 skipped.
- Observation: Even visually flat residuals retain stable device-specific structure when averaged statistically.

```
noise = patch.astype(np.float32) - denoised.astype(np.float32)  
# Visualized with: plt.imshow(noise, cmap='gray', vmin=-20, vmax=20)
```

1D Row Noise Pattern

Method: Average all rows of the noise residual along axis=0 to produce a 1D signal of length equal to the number of columns (768).

$$Pattern_{1D}(y) = \frac{1}{H} \sum_{x=1}^H Noise(x, y)$$

Mathematical Step: `row_pattern = np.mean(noise, axis=0)`

Plot (Index 20): The row noise pattern shows rapid fluctuations around zero, with scanner-specific column structure.

Observation: The signal is not visually clean, but its statistical pattern is repeatable across patches from the same scanner.

Stability: Averaging more patches makes the fixed component stronger while the random component cancels out.

```
def get_row_pattern(noise):  
    return np.mean(noise, axis=0) # shape: (768,)
```

Experiment 1 — 2D Reference Pattern (Baseline)

- Method: Average all training noise patches pixel-by-pixel per scanner to create a 2D fingerprint, then classify test patches via Pearson correlation.

Train/Test Split: Group-aware split with a separate held-out test set.

Overall Accuracy: 67.19%.

Interpretation: The 2D baseline works, but it is weaker than the feature-based SVM because the raw pixel fingerprint is sensitive to image content and local variation.

Accuracy: 0.6718857299821454

Classification Report:

	precision	recall	f1-score	support
s1_epson_4490	0.90	0.52	0.66	3110
s2_hp_scanjet_6300c_1_SG9C0270W5	0.36	0.53	0.43	577
s3_hp_scanjet_6300c_2	0.44	0.45	0.44	802
s4_hp_scanjet_8250	0.68	0.93	0.79	2792
accuracy			0.67	7281
macro avg	0.60	0.61	0.58	7281
weighted avg	0.72	0.67	0.67	7281

Confusion Matrix:

```
[[1629  312  318  851]
 [  42  307  108  120]
 [  75  140  364  223]
 [  67   88   45 2592]]
```

Experiment 2 — 1D Row Reference Pattern

- Method: Collapse each noise patch to a 1D column-wise average, build a per-scanner row reference, and classify using 1D correlation.

Train/Test Split: Same group-aware evaluation strategy used across the report.

Overall Accuracy: 70.31%.

Interpretation: The 1D row reference is stronger than the 2D baseline because it reduces scene dependence and emphasizes the scanner's stable column-wise signature.

Accuracy: 0.7030627661035572

Classification Report:

	precision	recall	f1-score	support
s1_epson_4490	0.91	0.57	0.70	3110
s2_hp_scanjet_6300c_1_SG9C0270W5	0.33	0.59	0.42	577
s3_hp_scanjet_6300c_2	0.46	0.60	0.52	802
s4_hp_scanjet_8250	0.77	0.91	0.83	2792
accuracy			0.70	7281
macro avg	0.62	0.66	0.62	7281
weighted avg	0.76	0.70	0.71	7281

Confusion Matrix:

```
[[1774  361  372  603]
 [  46  338  131   62]
 [  42  194  479   87]
 [  77  130   57 2528]]
```

Experiment 3 — 16-D Feature SVM Classifier

- Method: Extract 16 statistical features from each residual patch, standardize them, and train an RBF-kernel SVM with group-aware cross-validation.

Train/Test Split: GroupShuffleSplit for hold-out testing and GroupKFold for hyperparameter search.

Overall Accuracy: 88.44%.

Interpretation: This is the best-performing model in the project because the feature vector captures both row-pattern statistics and correlation structure.

```
=====
Test Accuracy : 0.8844
Best CV Score : 0.8840
Best Params   : {'C': 50, 'gamma': 0.1, 'kernel': 'rbf'}
=====

Classification Report:

              precision    recall  f1-score   support

s1_epson_4490      0.90      0.96      0.93      2111
s2_hp_scanjet_6300c_1_SG9C0270W5      0.57      0.48      0.52      362
s3_hp_scanjet_6300c_2      0.67      0.59      0.62      543
s4_hp_scanjet_8250      0.97      0.96      0.97      1838

   accuracy      0.88      4854
  macro avg      0.78      0.75      0.76      4854
weighted avg      0.88      0.88      0.88      4854


Confusion Matrix:

s1_epson_4490 \
s1_epson_4490      2033
s2_hp_scanjet_6300c_1_SG9C0270W5      71
s3_hp_scanjet_6300c_2      101
s4_hp_scanjet_8250      47

s2_hp_scanjet_6300c_1_SG9C0270W5 \
s1_epson_4490      18
s2_hp_scanjet_6300c_1_SG9C0270W5      172
s3_hp_scanjet_6300c_2      102
s4_hp_scanjet_8250      8

s3_hp_scanjet_6300c_2 \
s1_epson_4490      34
s2_hp_scanjet_6300c_1_SG9C0270W5      113
s3_hp_scanjet_6300c_2      318
s4_hp_scanjet_8250      13

s4_hp_scanjet_8250 \
s1_epson_4490      26
s2_hp_scanjet_6300c_1_SG9C0270W5      6
s3_hp_scanjet_6300c_2      22
s4_hp_scanjet_8250      1770

In [26]:
```

Results Comparison — All Three Experiments

- Experiment 1 (2D Correlation): 67.19% accuracy; weakest on the two HP 6300c units.
- Experiment 2 (1D Row Correlation): 70.31% accuracy; better than the 2D baseline and more robust to scene variation.
- Experiment 3 (16-D SVM Classifier): 88.44% accuracy; strongest and most balanced overall.
- Hardest Pair: s2 and s3, because both are HP ScanJet 6300c units and are very similar at the hardware level.
- Key Takeaway: The feature-based SVM is substantially better than direct correlation baselines on this real, imbalanced dataset.

Technical Stack

- PIL / Pillow: Safe loading of large TIFF files with `Image.MAX_IMAGE_PIXELS = None`.
- OpenCV: Grayscale image loading and patch handling where needed.
- PyWavelets: 4-level db8 wavelet denoising with soft thresholding.
- NumPy + SciPy: Residual computation, row averaging, correlations, and summary statistics.
- Scikit-learn: Label encoding, scaling, `GroupShuffleSplit`, `GroupKFold`, `GridSearchCV`, and SVM classification.
- Pandas / Matplotlib: Result logging, preview tables, and visualisation of patches, noise residuals, and feature plots.

References

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